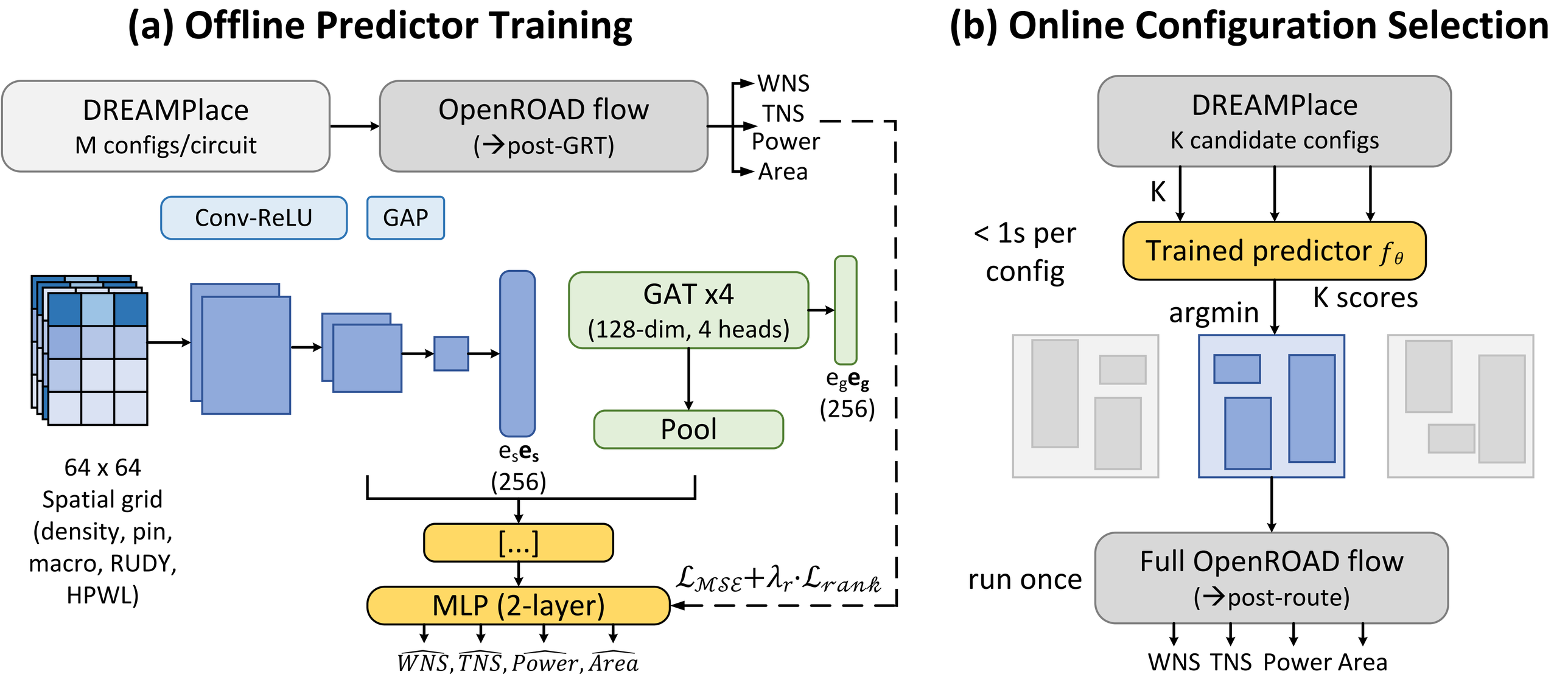
(a) Offline Predictor Training
DREAMPlace
M configs/circuit
OpenROAD flow
(→post-GRT)
WNS
TNS
Power
Area
Conv-ReLU
GAP
GAT x4
(128-dim, 4 heads)
Pool
$e_g \mathbf{e_g}$
(256)
$e_s \mathbf{e_s}$
(256)
64 x 64
Spatial grid
(density, pin,
macro, RUDY,
HPWL)
[...]
MLP (2-layer)
$\mathcal{L}_{MSE} + \lambda_r \cdot \mathcal{L}_{rank}$
$\widehat{WNS}, \widehat{TNS}, \widehat{Power}, \widehat{Area}$
(b) Online Configuration Selection
DREAMPlace
K candidate configs
K
< 1s per
config
Trained predictor f_θ
argmin
K scores
run once
Full OpenROAD flow
(→post-route)
WNS TNS Power Area